**House Prices: Advanced Regression Techniques — ML Assignment 1**
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This assignment was a clear regression problem as we were tasked with having to predict house prices using a dataset that contained 79 features besides the target variable. The assignment was assessed based on the RMSE metric on log prices. This immediately changes the approach as it is clear that I want to be careful with outliers as RMSE penalizes strongly on outliers. The whole model is trained on the house prices after the log transformation which I understand is the best approach to this problem.

I found that in the dataset there are 4 values which I classified as outliers specifically due to their large area in square feet. Two seemed to be outliers which followed the pattern of the data, while the other two were sold for an extremely low price. I assume that this may be caused by other variables such as location, how old the houses are, or even the owner's need for money at the moment. In any case, I made the decision to remove them as they could harshly affect the model's training. After making various failed models, I went back and made sure to only remove these data points from the training data which forced me to create a training pipeline which was robust when using Kfold validation. When removing these outliers, I improved the model by a 0.0168 which was my most significant gain in the model.

As we learned in class, I decided to use PyTorch to create a neural network as my main model and compared it against a dummy model which only predicted the mean. In my neural network I trained on 253 features which emerged after one-hot encoding multiple categorical variables in order for them to be efficiently used in the model without giving them numerical values that would create inherent ordering of categories which are non-existent in the data. I chose to create my model using batches for gradient descent, ReLu activation for faster learning, and a dropout with a factor of 0.4 in order to avoid overfitting in my model. After training the model I performed feature engineering which was mostly insignificant (besides fixing the skewed data) along with hyperparameter tuning, and finally decided to average the models with 5 different seeds in order to get the best model possible from this combination.

**Results table**

| Change | CV RMSE | Gain |
|---|---|---|
| Baseline (predict mean) | 0.3991 | — |
| Neural network, all data | 0.1535 | — |
| − outliers from training | 0.1367 | 0.017 |
| + skew correction | 0.1334 | 0.003 |
| + derived features | 0.1323 | 0.001 (noise) |
| + tuning | 0.1314 | 0.001 (n.s.) |
| + 5-seed averaging | **0.1280** | 0.003 |
| **Kaggle public** | **0.12900** | |

As you can see, I did continuously improve the model with all my corrections, except with the feature engineering and hyperparameter tuning which had an improvement so small it is not significant enough to assume it was an actual improvement. It is important to note that my model predicted a 0.1280 CV RSME and Kaggle gave me a 0.12900. This is a good sign as it shows that my model was robust and did not overfit.

Unfortunately my model has limitations I want to address with deciding not to clip my final predictions as my main limitation. After reviewing everything, this was suggested to me by Claude as I had an extreme value of a house being predicted to sell at $1.23 million, but I decided this was not something I would do as the training dataset also had extreme outliers and I didn't want the possibility of this happening in the future data to reduce my model. Additionally, I also believe that using a neural network for this assignment is not the best approach as there are better options for these regressions such as XGBoost.